

CSC1110 Lab 1: Basic Text Processing and Language Models

Recap of Unigram Language Models

Word Probability

In order to calculate the empirical probability of a word (i.e. the probability of a word based on some data/corpus), we use the following formula:

$$P(word) = \frac{Count(word)}{n} \quad (1)$$

where n is the number of words in a corpus.

Unigram Model

Unigram models calculate the probability of a given sentence based on the following formula:

$$P(w) = \prod_i^n P(w_i) \quad (2)$$

where $P(w)$ is the empirical probability of a sentence w with n words and w_i is a word.

Below is the same formula with slightly simpler notation:

$$P(w) = P(w_0) \times P(w_2) \times \dots \times P(w_n) \quad (3)$$

Questions

1. Load corpus “AcademyAwardsWikipedia.txt”
2. Write pre-processing code that:
 - (a) Remove punctuation with a regular expression (hint: regex substitution could be useful here, e.g. replace anything that is not alphanumeric or a hyphen with an empty string. In Python, you might use `re.sub`)

- (b) Converts words to lowercase.
 - (c) Performs tokenisation by splitting the data based on whitespace and new-line characters.
3. Build a unigram model for the corpus above using only your programming language of choice standard libraries/ external mathematical libraries (e.g. NumPy in Python).
 4. Build a unigram model for the corpus above using only standard programming libraries and mathematical/array libraries (e.g. NumPy).
 5. What is the sentence-level probability for the sentence "The best actor" based on your model?
 6. What is the sentence-level probability for "The best actor in Ireland" based on your model?

*** The solutions will be made available after the lab, before the Monday lecture.**